Vizora

Automated Photo Identification and Classification

for Traffic Violations Using Computer Vision

Problem Statement 3

Flipkart Gridlock 2.0 Hackathon - Round 2

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1. Executive Summary

Vizora proposes a privacy-focused, locally deployable computer vision system for automated traffic violation identification from surveillance imagery. The system ingests photos, short clips, or near-real-time camera feeds; detects vehicles, riders, pedestrians, and plates; classifies traffic violations; extracts number plates; and generates tamper-evident evidence packets for review and enforcement.

The design keeps the core inference pipeline independent of external API providers. Detection, OCR, violation reasoning, evidence generation, and review workflows are designed to run inside the deployment boundary using self-hosted services, PostgreSQL, Redis, and MinIO/S3-compatible storage.

Hackathon fit

- Directly addresses automated photo identification and classification for traffic violations using computer vision.
- Starts with still-image upload and batch processing, then extends to temporal bursts and near-real-time feeds.
- Produces reviewer-friendly, court-oriented evidence packets instead of only raw model labels.
- Includes privacy safeguards, model optimization, organization roles, live camera handling, human review, and an evaluation plan suitable for deployment discussion.

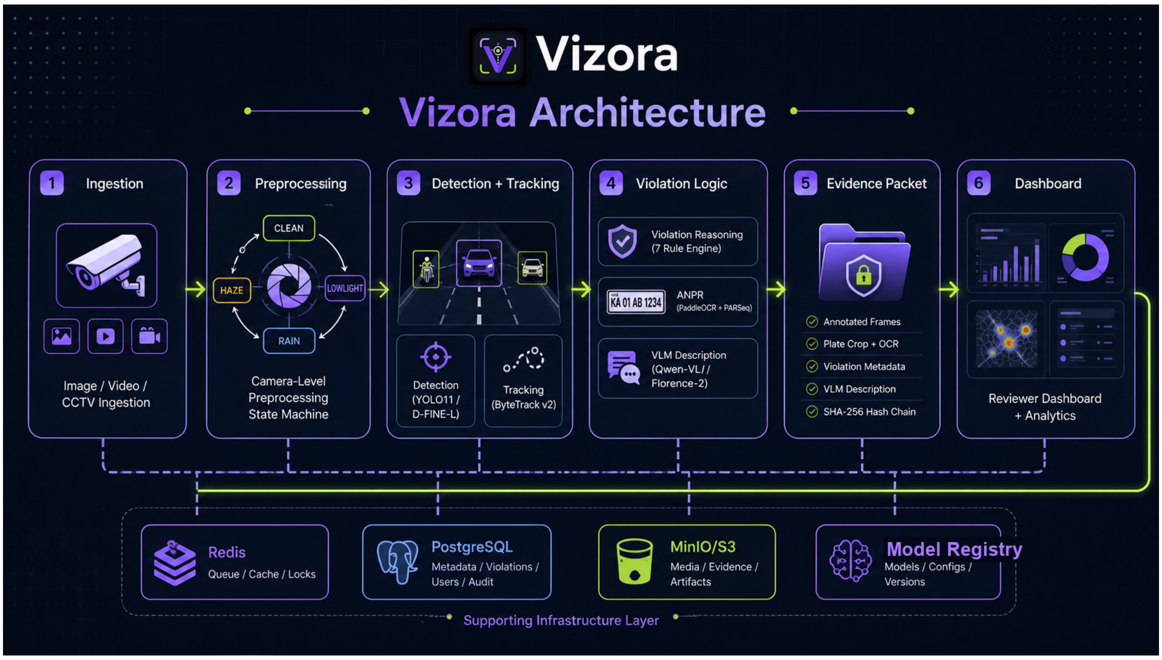
2. Problem Understanding

Traffic surveillance systems generate a large volume of imagery, but manual review is slow, inconsistent, expensive, and difficult to scale across intersections and cities. A practical solution must detect traffic participants, understand scene context, identify violations, read plates when visible, and produce evidence that can be audited by human reviewers.

The key technical challenge is that several violations are not visible from object detection alone. Helmet and seatbelt violations may be inferred from a single frame, but wrong-side driving, red-light crossing, stop-line violation, illegal parking, and robust triple-riding decisions usually require temporal context, scene geometry, or signal phase metadata. Vizora therefore treats single-image inference as the baseline path and temporal reasoning as an accuracy and deployability enhancer.

3. Proposed Approach

The proposed approach is an end-to-end pipeline with four major layers: ingestion, vision inference, evidence generation, and reviewer analytics. Each stage is modular so that models can be changed through a validated registry instead of hardcoded pipeline choices.



Core pipeline stages

- Ingestion: upload a photo, batch of images, short clip, or feed frames from a camera source.
- Preprocessing: apply consistent camera-level enhancement using a sticky CLEAN, LOWLIGHT, HAZE, RAIN, or MULTI mode.

- Detection and tracking: detect vehicles, riders, pedestrians, helmets, and plates; attach track IDs when temporal input is available.
- Violation reasoning: combine detections with scene configuration such as stop lines, lane direction vectors, no-parking zones, and signal phase.
- ANPR: detect plate crop, run PaddleOCR first, then super-resolution plus PARSeq fallback for low-confidence results.
- Evidence generation: create annotated frames, metadata JSON, hash-chain records, and a natural-language explanation from the VLM module.
- Dashboard: support reviewer workflows, analytics, cameras, evidence viewer, live feed events, and low-confidence review queue.

4. Organization, Roles, and Access Model

Vizora is structured as a multi-tenant platform for traffic enforcement organizations. The organization is the top-level boundary: users, cameras, scene configurations, violations, evidence packets, plate hashes, review decisions, and analytics are all scoped by org_id.

Role	Responsibility
Admin	Creates the organization, manages users, registers cameras, configures scenes, and can approve or reject violations.
Reviewer	Reviews evidence packets, corrects OCR when needed, and approves, rejects, or escalates violations.
Analyst	Views trends, hotspot maps, repeat-offender summaries, camera performance, and evaluation dashboards.
Operator	Monitors live camera status, preprocessing mode, stream health, queue depth, and camera downtime.

Setup flow

An organization is registered first, the first user becomes the admin, and the admin invites reviewers, analysts, and operators. JWT authentication identifies the user and organization for every API call. This keeps one organization's cameras, evidence, plate hashes, and reviewer actions isolated from all others while still allowing city-scale multi-camera operations.

5. Live Footage and Camera Stream Handling

Live footage is handled through camera source records rather than raw browser streaming. Each camera stores its organization, GPS/location, stream source, scene geometry, preprocessing mode, health status, target FPS, and queue metadata. The backend samples frames and sends events to the dashboard after processing, so the UI remains an evidence and review surface rather than a CCTV video player.

Source type	How Vizora uses it
RTSP stream	Primary CCTV/IP camera path. A worker pulls frames with OpenCV, GStreamer, or FFmpeg, samples at configured FPS, and places frames into Redis for inference.
HTTP snapshot	Polls JPEG snapshots for cameras that do not expose RTSP. Useful for low-cost or legacy cameras.
Uploaded short clip	Demo-friendly path that exercises the same temporal pipeline without requiring physical camera access.
File watcher / batch folder	Offline testing path for surveillance image batches and evaluation datasets.

Near-real-time processing flow

Camera source -> frame sampler -> Redis queue -> preprocessing state machine -> detection/tracking -> violation reasoning -> evidence packet -> PostgreSQL + MinIO -> SSE event -> dashboard feed/review queue.

Each live camera keeps a sticky preprocessing mode, so all frames from that camera remain visually consistent for ByteTrack. If frames drop or the camera goes offline, health status changes and active tracks are closed cleanly rather than producing unreliable violation records.

6. Methodology

6.1 Image-first processing

Because the problem statement emphasizes photo identification, the base path accepts a single uploaded image and returns detections, violation candidates, OCR output, confidence, and a review recommendation. Temporal features are optional enhancements for short clips and live feeds. This prevents still-image use cases from depending on ByteTrack or multi-frame data.

6.2 Camera-level preprocessing state machine

Preprocessing is modeled as a per-camera state machine rather than a per-frame switch. Each camera maintains a rolling quality state and applies a consistent mode to all frames until sustained evidence supports a transition. This avoids enhancement flicker, stabilizes tracking, and keeps the pipeline latency deterministic.

6.3 Violation reasoning

Violation	Primary method	Input confidence
Helmet	Rider/person detection plus head ROI helmet classifier	Single image
Seatbelt	Driver window crop plus binary classifier	Single image
Triple riding	Person count associated with motorcycle track	Still image or temporal confirmation
Wrong-side driving	Track trajectory vector compared with lane direction vector	Temporal
Stop-line violation	Vehicle box or track crosses configured stop-line polygon	Temporal or review in still image
Red-light violation	Stop-line crossing during red signal phase	Temporal plus signal metadata
Illegal parking	Stationary track inside no-parking polygon for threshold duration	Temporal

6.4 Evidence packet generation

Every confirmed or review-worthy violation produces an evidence packet containing annotated image frames, violation metadata, plate OCR output, confidence scores, timestamps, camera ID, GPS or scene reference, model profile, and a SHA-256 hash chain. Public-facing copies apply face blur and plate redaction, while sealed review copies preserve required evidence under access control.

7. Model Strategy and Optimization

Vizora uses a Pydantic-validated model registry so the same pipeline can run in fast, accuracy, edge, and review profiles. This lets the team balance latency, GPU availability, and accuracy without rewriting the pipeline.

Stage	Default / proposed model	Reason
Detector	YOLO11 for MVP, D-FINE-L for accuracy profile	Fast tooling first, stronger accuracy profile when integration time allows
Tracking	ByteTrack v2	Stable IDs and low-confidence association for occluded vehicles
Pose	RTMO-m	Single-stage multi-person pose for rider counting and head ROI extraction
Classifiers	EfficientNetV2-S	Efficient binary classification for helmet and seatbelt tasks
ANPR	YOLO11-plate + PaddleOCR PP-OCRv5 + PARSeq fallback	Regional plate support first, fallback for dirty or stylized crops
VLM	Qwen-VL preferred, Florence-2 fallback	Explanation only; it never overrides OCR or violation logic

Optimization choices

- TensorRT FP16 serving path for GPU deployments.
- Hard 15 ms preprocessing budget per frame; heavy enhancement is dropped, deferred, or routed to review rather than destabilizing latency.
- Crop-level expensive work: Real-ESRGAN and PARSeq are used on plate crops or review frames, not every full frame.
- Profile-based model registry supports fast, accuracy, edge, and review modes.
- Asynchronous queues decouple ingestion, inference, VLM description generation, and evidence storage.

8. Privacy and Security Approach

The system is privacy-focused by design. Core inference does not rely on external API providers, and sensitive evidence can remain inside the organization or city deployment boundary. Storage uses PostgreSQL for structured records and MinIO/S3-compatible object storage for evidence files. Redis is used for queueing and transient workflow state.

- Face blur and plate redaction for public-facing evidence copies.
- Plate hashing for lookup and repeat-offender analytics without exposing raw plate values broadly.

- Role-based reviewer actions and evidence status tracking in the dashboard.
- Tamper-evident SHA-256 hash chain across evidence packet frames and metadata.
- Human review for low-confidence, ambiguous, or legally sensitive cases.

9. Assumptions

Area	Assumption	Impact
Organization	Traffic teams operate inside organization-scoped workspaces with assigned roles.	Evidence, cameras, users, and analytics remain isolated by org_id.
Camera scene	Each deployed camera has stop-line polygons, lane direction vectors, no-parking zones, and optional signal ROI metadata.	Temporal violation logic becomes explainable and auditable.
Live streams	RTSP, HTTP snapshots, uploaded clips, or file batches are available for demo and deployment.	Same inference pipeline can support both photo-first and live modes.
Input quality	Images may include night, rain, fog, glare, blur, occlusion, dense traffic, and partial vehicles.	Preprocessing and review workflow must handle degradation.
Plate visibility	Not every frame contains a readable plate.	ANPR confidence controls fallback and human review.
Legal context	Emergency vehicles, child-on-lap cases, already-past-stop-line cases, and turban/pagdi riders require special handling.	Rules include exemptions and review routing.

10. Expected Evaluation Strategy

The evaluation plan measures not only model accuracy, but also OCR quality, latency, robustness, evidence trust, and human review effectiveness. This is important because enforcement systems fail when they are accurate in isolation but unreliable under weather, occlusion, or ambiguous legal conditions.

Category	Metrics	Target evidence
Object detection	mAP@0.5, mAP@0.5:0.95, per-class AP	Vehicle, rider, pedestrian, plate detection report
Violation classification	Precision, recall, F1, confusion matrix, Brier score	Seven violation categories with calibrated confidence
ANPR	Plate detection recall, character accuracy, full-plate exact match	Plate crop benchmark with low-confidence fallback tracking
Tracking/live stream	IDF1, ID switches, stream uptime, frame drop rate, queue latency	RTSP/clip tests for temporal violations and live feed events
Latency	p50 and p95 preprocessing, inference, OCR, total response time	Single image, batch, and near-real-time feed tests
Robustness	Accuracy by day/night, rain/clear, fog/clear, dense/sparse traffic	Stratified dataset slices and edge-case test set
Trust and review	Hash verification pass rate, review queue precision, duplicate cooldown behavior	Evidence packet audit and reviewer workflow test
Access control	Role permission checks and org isolation checks	Admin/reviewer/analyst/operator scenario tests

11. Implementation and Deliverables

The implementation is organized as a FastAPI backend for processing and evidence APIs, a Next.js dashboard for reviewers and judges, PostgreSQL for structured records, Redis for queues, and MinIO/S3-compatible object storage for evidence images and crops.

- MVP: image upload, detector output, visible violation classification, plate OCR, annotated result, evidence metadata.
- Organization setup: org registration, first admin, invited users, role-scoped dashboards, and org-scoped API access.
- Live feed extension: camera registration, RTSP/HTTP/clip/file sources, frame sampling, SSE events, camera health, and queue depth monitoring.
- Temporal extension: burst or feed processing, ByteTrack IDs, wrong-side, red-light, stop-line, illegal parking logic.
- Review workflow: low-confidence routing, approve/reject/escalate actions, evidence viewer, OCR correction.
- Analytics: KPIs, violation distribution, hotspot map, repeat-offender table, time-of-day trends.

12. Risks and Mitigations

Risk	Mitigation
Still images cannot prove all temporal violations.	Return lower confidence, require scene metadata, and route ambiguous cases to review; use short bursts where available.
Live streams drop frames or disconnect.	Track camera health, sample at configured FPS, close stale tracks safely, and show operator-visible status.
Role mistakes expose sensitive evidence.	Scope all records by organization and enforce role checks on review, camera, analytics, and evidence actions.
Weather and lighting degrade detection.	Use camera-level preprocessing state and crop-level fallback for expensive enhancement.
OCR errors on dirty or regional plates.	Use PaddleOCR primary, PARSeq fallback after super-resolution, regex validation, and human review.
False positives on legally exempt cases.	Encode edge-case rules for turbans, emergency vehicles, child-on-lap nuance, and already-past-stop-line cases.

13. Conclusion

Vizora is designed as a practical, privacy-first traffic violation intelligence system. It starts with automated photo identification, then extends into organization-scoped operations, live camera processing, temporal reasoning, evidence integrity, and reviewer workflows. The central design principle is to make every output explainable, auditable, and deployable: models propose detections, rule logic validates violations, OCR is confidence-gated, VLM output describes evidence only, and humans remain in the loop for ambiguous cases.